Day 1 – Environment Setup & Initial Hand Tracking

1. Day Information

Day Number: 01
Date: 15/01/2026
Time Spent: ~4–5 hours

2. Objective of the Day

Set up a working Python environment and successfully run a real-time hand landmark detection script as the foundation for Amharic Sign Language recognition.

3. Tasks Performed

• Installed multiple Python versions
• Attempted MediaPipe installation
• Created and activated virtual environment
• Installed OpenCV and MediaPipe
• Configured VS Code interpreter
• Ran hand landmark test script successfully

4. Technical Problems Encountered

• ModuleNotFoundError: mediapipe
• AttributeError: mediapipe has no attribute 'solutions'
• No module named cv2
• Python version mismatch between CMD and VS Code
• Virtual environment not being used by VS Code

5. Root Cause Analysis

• MediaPipe not compatible with Python 3.13
• Packages installed outside active virtual environment
• VS Code defaulted to global Python interpreter

6. Solution & Fix Implemented

• Installed Python 3.12.2
• Set correct Python interpreter in VS Code
• Recreated virtual environment
• Installed dependencies inside venv
• Verified setup by running webcam hand landmark script

7. Lessons Learned

• Always verify Python compatibility before ML installs
• Virtual environments are mandatory for ML projects
• Interpreter selection directly affects execution

8. What to Avoid Next Time

• Avoid experimental Python versions
• Avoid installing packages globally
• Avoid assuming VS Code uses the correct interpreter

9. Next Day Plan

• Implement landmark data saving
• Label common Amharic gestures
• Begin dataset construction